

Cryptography - Exam Preparation.

Problem 1

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
P	Q	R	S	T	U	V	W	X	Y	Z				
15	16	17	18	19	20	21	22	23	24	25	26			

SHIFT CIPHER

Shift cipher $\rightarrow$ Encryption: $x \rightarrow (x+b) \bmod 27$

Decryption: $x \rightarrow (x-b) \bmod 27$

We know: $K \rightarrow F : 10 \rightarrow 5$

So $b = -5 \equiv 22 \bmod 27$

Ciphertext: N L Y M L O V R M D O D I B

13 26 24 12 26 20 21 17 12 3 14 3 8 1

Plaintext: 18 4 2 17 4 25 26 22 17 8 19 8 13 6

S E C R E T L W R I T I N G

Problem 2:

2 dimensional Hill-cipher over alphabet 27.

Plaintext: SURE $\rightarrow$ 18 20 17 4

HILL CIPHER

$$\begin{bmatrix} 1 & 8 \\ 11 & 9 \end{bmatrix} \begin{bmatrix} 18 \\ 20 \end{bmatrix} = \begin{bmatrix} 178 \\ 378 \end{bmatrix} \equiv \begin{bmatrix} 16 \\ 0 \end{bmatrix} \bmod 27 \quad Q$$

$$\begin{bmatrix} 1 & 8 \\ 11 & 9 \end{bmatrix} \begin{bmatrix} 17 \\ 4 \end{bmatrix} = \begin{bmatrix} 49 \\ 223 \end{bmatrix} \equiv \begin{bmatrix} 22 \\ 7 \end{bmatrix} \bmod 27 \quad W$$

Ciphertext: Q A W H

Problem 3:

Affine cipher: Encryption using a key (a,b) :

Encryption: $x \rightarrow (ax+b) \bmod 27$

Decryption: $x \rightarrow (a\bar{a} - \bar{a}b) \bmod 27$

where $\bar{a} \in \mathbb{Z}_{27}^*$ is the modular inverse of $a \bmod 27$.

AFFINE CIPHER

$a \in \mathbb{Z}_{27}^*$, a needs to be invertible mod 27.

Here $(a, b) = (2, 3)$ so $\bar{a} = 14$

Decryption is $x \rightarrow (14x + 12)$

(Note: $-\bar{a}b = -14(3) = -42 \equiv 12 \pmod{27}$)

Ciphertext: R T J J L C
17 19 9 9 11 2

Plaintext: 7 8 3 3 4 13
H I D D E N

Problem 4:

Affine cipher, find (a, b)

We know $D \rightarrow W$ and $L \rightarrow F$

So $3 \rightarrow 22$ and $11 \rightarrow 5$

So $8a + b = 22$ and $11a + b = 5$

Hence: $8a = 5 - 22 \equiv 10 \pmod{27}$

$\gcd(8, 27) = 1$ $\bar{8} = 17 \pmod{27}$

Indeed, $\varphi(27) = \varphi(3^3) = 27 - 3 = 24$ so $\bar{8} = 8^{23} \pmod{27}$

Hence: $a \equiv 17 \cdot 10 \equiv 8 \pmod{27}$

$3 \cdot 8 + b = 22 \Rightarrow b = -2 = 25 \pmod{27}$

So $(a, b) = (8, 25)$

Problem 5:

Define what is meant by a modular inverse:

A modular inverse of $a \in \mathbb{Z}$ modulo n is any integer b with $ab \equiv 1 \pmod{n}$.

Is 3 invertible modulo 1234?

Check for divisibility: not divisible by 3 so $\gcd(3, 1234) = 1$

So 3 is invertible modulo 1234 $3 \in \mathbb{Z}_{1234}^*$

Problem 6:

Is $\begin{bmatrix} 23 & 42 \\ 45 & 27 \end{bmatrix}$ invertible over $\mathbb{Z}_{68}$?

68 136 204 272 340 408 476 544 612 680
 748 816 884 952 1020

$$\det = [23(57) - 42(45)] = -579 \equiv -35 \equiv 33$$

$$\begin{array}{c|cc} 1 & 23 & 42 \\ \hline 33 & 45 & 27 \end{array} \Rightarrow \text{inverse of } 33 \text{ mod } 68 = 33$$

$$33 \begin{bmatrix} 27 & -42 \\ -45 & 23 \end{bmatrix} = \begin{bmatrix} 891 & -1386 \\ -1485 & 759 \end{bmatrix} \equiv \begin{bmatrix} 61 & -42 \\ -11 & 57 \end{bmatrix} \equiv \begin{bmatrix} 61 & 26 \\ 57 & 57 \end{bmatrix}$$

Problem 7:

Explain in which sense Hill ciphers and Vigenere ciphers are both special cases of affine ciphers.

Vigenere cipher: This is when $a = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \text{Id}$, b is arbitrary.

So encryption is $x \rightarrow x + b$ and decryption is $x \rightarrow x - b$.

Hill cipher: This is when $a \in \text{GL}_d(\mathbb{Z}_n)$, $b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ so

encryption is $x \rightarrow Ax$ and decryption $y \rightarrow \bar{A}y$.

where $A \cdot \bar{A} = \text{Id}$ over $\mathbb{Z}_n$.

Problem 8:

Explain what is meant by letter frequency analysis and give an example of a cryptosystem vulnerable to it.

Letter frequency analysis is the process of analysing the statistical properties of letters and how frequently they occur. For example, the most frequently used letters in the English language are e, t, a, ...

The statistical properties of letters are still intact after encryption and so this can be used to decrypt the message.

- Affine ciphers are vulnerable - Vigenere, Hill ciphers.
- Shift ciphers

Problem 9:

Describe what is meant by a brute force attack and describe a scenario where this is feasible.

Brute force attack is when an attacker tries out all possible keys possible to try and decrypt the encrypted message. This is a trial and error method of attacking a cryptosystem.

Shift ciphers over a small alphabet (26) are vulnerable to brute force attacks as there is only 26 possible keys.

Problem 10:

- (a) The key space of an n letter alphabet for a shift cipher = n .
The key space for a general substitution cipher = $n!$
- (b) Yes I agree with Bob, General substitution ciphers are not as vulnerable to brute force attackers.
Taking alphabet 26, the key space is 4×10^{26} which would take much longer to break than having only 26 possibilities.

Problem 11:

What is meant by a (symmetric) cryptosystem.

A cryptosystem consists of:

- A set $\mathcal{P}$ of plaintexts
- A set $\mathcal{C}$ of ciphertexts
- A set $\mathcal{K}$ of keys (the key space).

For each key $k \in \mathcal{K}$ functions $e_k: \mathcal{P} \rightarrow \mathcal{C}$ and $d_k: \mathcal{C} \rightarrow \mathcal{P}$ such that $d_k(e_k(x)) = x$ for all $x \in \mathcal{P}$.

In other words: for each plaintext x , if we encrypt x using a key k to obtain a ciphertext $y = e_k(x)$ we may decrypt y to recover $x = d_k(y)$.

Problem 12:

Kerckhoffs' Principle:

When we study the security of a cryptosystem we assume that an adversary knows exactly which cryptosystem is being used but that (s)he does not have the key being used. In this way, the overall security depends only on the mathematical properties of the cryptosystem and the plaintexts and the keys used rather than our ability to keep the encryption / decryption algorithms secrets.

Problem 13:

What is meant by a side-channel attack?

This involves decrypting a message using information related to the practical implementation of the cryptosystem.

Eg. monitoring the power consumption during encryption to learn something about the keys being used.

Problem 14:

What does it mean for a cryptosystem to be perfectly secure?

Our cryptosystem is perfectly secure if $\text{Prob}(x=y \mid Y=y) = \text{Prob}(X=x)$ for all $x \in \mathcal{P}$ and $y \in \mathcal{C}$.

In other words, perfect security means that knowing the ciphertext doesn't tell us anything about the corresponding plaintext.

Problem 15:

State Shannon's Theorem on perfectly secure cryptosystems.

Shannon's Theorem: Suppose that $|\mathcal{P}| = |\mathcal{C}| = |\mathcal{K}|$. Then our cryptosystem is perfectly secure if and only if:

- $\text{Prob}(K=k) = \frac{1}{|\mathcal{K}|}$ for all $k \in \mathcal{K}$
- $\forall x \in \mathcal{P}, \forall y \in \mathcal{C} \exists k \in \mathcal{K} \text{ s.t. } \text{ek}(x)=y.$

Problem 16:

show that $|\mathcal{K}| \geq |\mathcal{C}| \geq |\mathcal{P}|$

- Pick $k \in \mathcal{K}$ since $\text{dec}(\text{ek}(x)) = x$ for all $x \in \mathcal{P}$, ek is injective from $\mathcal{P} \rightarrow \mathcal{C}$. Hence $|\mathcal{P}| \leq |\mathcal{C}|$. (Note: this did not use perfect security).
- let $x \in \mathcal{P}$ and $y \in \mathcal{C}$ then: $\text{Prob}(Y=y \mid X=x) = \text{Prob}(Y=y) > 0$. In particular: $\forall x \in \mathcal{P}, \forall y \in \mathcal{C}, \exists k \in \mathcal{K}, y = \text{ek}(x)$. Hence: given any $x \in \mathcal{P}$, the function $\mathcal{K} \rightarrow \mathcal{C}, k \mapsto \text{ek}(x)$ is surjective $\leadsto |\mathcal{K}| \geq |\mathcal{C}|$.

Problem 17:

Describe what is meant by a one-time pad:

Let $\mathcal{P} = \mathcal{C} = \mathcal{K} = \mathbb{Z}_n^d$ (any values of d, n) with $e_k(x) = x+k$, $d_k(y) = y-k$ and uniformly chosen keys. By Shannon's theorem this is then perfectly secure: condition (a) is satisfied by construction, for $x, y \in \mathbb{Z}_n^d$, the unique $k \in \mathbb{Z}_n^d$ with $x+k=y$ is $k=y-x$.

- Advantage: Perfectly secure cryptosystem.
- Disadvantage: We struggle to securely exchange a key that is as large as the plaintexts and perfect security requires keys never to be reused!

Problem 18:

Describe Block ciphers and stream ciphers.

Block ciphers operate on blocks of ^{plain}text, and ciphertext symbols to produce blocks of plaintext and ciphertext symbols respectively.

Example: Affine ciphers. DES (Data Encryption Standard).

Stream ciphers: operate on streams of symbols over an alphabet. Each plaintext symbol in the input stream is encrypted in a way that depends on the previous history of the stream.

- Plaintexts are streams $a_1, a_2, \dots$ over $\mathbb{Z}_n$.
- Keys are seeds used to generate pseudo-random key streams $k_1, k_2, \dots$.
- Encryption: $a_1, a_2, \dots \rightsquigarrow a_1 + k_1, a_2 + k_2, \dots$.
- Decryption: subtraction.

Example: Synchronous / Asynchronous stream ciphers

eg AS/1 used in the GSM (Global System for Mobile communication).

Problem 19:

not LFSR definition

Explain what is meant by a linear feedback shift register:

A feedback shift register of length L is a particular type of machine.

The state of the register is a bit string $s_{L-1}, \dots, s_0$ of length L .

Given an initial state, the operation of the register is governed by a clock. During each clock cycle,

- 1) the current value of s_0 is appended to the output stream and

feedback
shift
register

- 11) The current state $s_{L-1}, \dots, s_0$ is then replaced by $f(s_{L-1}, \dots, s_0)$, $s_{L-1}, \dots, s_1$, where $f: \mathbb{Z}_2^L \rightarrow \mathbb{Z}_2^L$ is the feedback function of the register.

Problem 20:

definition for LFSR

A linear feedback shift register (LFSR) is a feedback shift register with feedback function of the form $f(s_{j-1}, \dots, s_{j-L}) = c_1 s_{j-1} + \dots + c_L s_{j-L}$ for fixed $c_1, \dots, c_L \in \mathbb{Z}_2$.

The connection polynomial of the above LFSR is:

$$C(x) = 1 + c_1 x + \dots + c_L x^L \text{ a polynomial over } \mathbb{Z}_2.$$

LFSR's and their connection polynomials determine each other.

Problem 21:

Define what is meant by a non-singular LFSR.

An LFSR of length L with connection polynomial $C(x)$ is non-singular if $\deg C(x) = L$ and singular if $\deg C(x) < L$.

Non singular LFSR's are purely periodic rather than mainly ultimately periodic: $\exists N \in \{1, \dots, 2^L - 1\} \forall j \geq 0, s_j = s_{j+N}$.

Problem 22:

Explain what is meant by a primitive polynomial.

Primitive polynomials are irreducible. Let $f(x)$ be irreducible of degree n over $\mathbb{Z}_2$. Then $f(x)$ is primitive if $x + f(x)$ generates the unit group of the field $\mathbb{Z}_2(x)/f(x) \cong \mathbb{F}_{2^n}$.

Problem 23:

Suppose that the attacker knows $2L$ consecutive bits of plaintext $x_0, \dots, x_{2L-1}$ and also the corresponding bits $y_0, \dots, y_{2L-1}$ of ciphertext. They then obtain $2L$ consecutive bits $s_0, \dots, s_{2L-1}$ from the key stream, where $s_j = x_j + y_j$. They wish to find the connection polynomial:

$$C(x) = 1 + c_1 x + \dots + c_L x^L$$

Note:

$$\begin{cases} S_L = C_1 S_{L-1} + \dots + C_L S_0 \\ S_{L+1} = C_1 S_L + \dots + C_L S_1 \\ \vdots \\ S_{2L-1} = C_1 S_{2L-2} + \dots + C_L S_{L-1} \end{cases}$$

This is a system of linear equations in L variables ($C_1, \dots, C_L$).
In practice, an attacker can probably uniquely solve this and then recover the LFSR used.

Problem 24:

Determine the least period of the resulting infinite sequence $s_0, s_1, \dots$

$$p(x) = 1 + x^2 + x^3.$$

t	$s_{t+2}^{x^2}$	$s_{t+1}^{x^1}$	$s_t^{x^0}$	these are added.
0	1	1	1	
1	0	1	1	
2	0	0	1	
3	1	0	0	
4	0	1	0	
5	1	0	1	
6	1	1	0	
7	1	1	1	initial state $\rightarrow$ period is 7

Problem 25:

"For symmetric cryptosystems, key distribution scales badly as the number of parties grows"

Suppose that a group of $n \geq 2$ people, any two wish to be able to securely communicate with each other. Then $\binom{n}{2} = \frac{n(n-1)}{2}$ keys need to be exchanged. Eg. to allow any two of the $\approx 17,000$ shareholders of NUG to communicate securely $\approx \binom{17000}{2}$ keys would need to be exchanged, and generalised. This is infeasible!

We could use a centralised key exchange server but then we would need to trust whoever is running that server!

This is where public key cryptography enters.

Problem 26:

Define public-key cryptography.

A public-key cryptosystem consists of sets $\mathcal{P}$, $\mathcal{C}$, $\mathcal{K}$ as before and functions $\text{enc}_K: \mathcal{P} \rightarrow \mathcal{C}$ and $\text{dec}_K: \mathcal{C} \rightarrow \mathcal{P}$ for $K \in \mathcal{K}$ with the following properties

- For each (public key) $e \in \mathcal{K}$, there exists (a private key) $d \in \mathcal{K}$ with $\text{dec}_d(\text{enc}_e(x)) = x$ for all $x \in \mathcal{P}$.

Problem 27:

Describe the key generation step, the encryption step, and the decryption step of the RSA cryptosystem.

- Key Generation:

Alice picks two different, large primes p and q and computes their product $n = p \cdot q$. She then chooses a random integer e with $1 < e < (p-1)(q-1)$ and $\gcd(e, (p-1)(q-1)) = 1$.

Her public key is (n, e) .

Next using the Euclidean Algorithm she computes the unique integer d with $1 < d < (p-1)(q-1)$ and $de = 1 \pmod{(p-1)(q-1)}$.

In other words, $d \in \mathbb{Z}_{(p-1)(q-1)}^\times$ is the inverse of e modulo $(p-1)(q-1)$. Alice's private key is (n, d) .

- Encryption:

Given a message $m \in \mathbb{Z}_n$, Bob uses Alice's public key (n, e) to compute the ciphertext $c = m^e \pmod n$.

- Decryption:

After receiving c , Alice uses her private key (n, d) to decrypt $m = c^d \pmod n$.

$$2, 3, 5, 7, 11, 13, 17, 19 \quad \gcd(14, 48) \neq 1$$

$$\gcd(e, (p-1)(q-1)) \neq 1$$

Problem 28:

Two random primes $p = 13$, $q = 5$ $p \cdot q = 65 = n$.

$$(p-1)(q-1) = 12 \cdot 4 = 48$$

$$de = 1 \pmod{48} \rightarrow d(14) = 1 \pmod{48} \rightarrow d = \frac{1}{14} \pmod{48}$$

Problem 29:

"If there is an efficient algorithm for integer factorization, then RSA is insecure."

The security of RSA depends on the difficulty of integer factorization. Let (n, e) be Alice's public key. Suppose that Oscar managed to factor $n = pq$ into a product of primes p and q . He can then duplicate the relevant parts of the key generation step and derive Alice's private key.

Therefore if there is an efficient algorithm for integer factorization for large numbers (trial division and is suitable for smaller numbers) RSA would be insecure.

Problem 30:

Define Euler's Totient

Let $n \geq 1$ and let $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$. Then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Proof: Let $R = \{r_1, \dots, r_{\varphi(n)}\}$ be a RRS mod n .

Exercise: Since $\gcd(a, n) = 1$, $R' := \{ar_1, \dots, ar_{\varphi(n)}\}$ is also a RRS mod n .

Hence: $\exists$ bijection $f: \{1, \dots, \varphi(n)\} \rightarrow \{1, \dots, \varphi(n)\}$ such that

$$ar_i \equiv r_{f(i)} \pmod{n} \quad \text{For all } i = 1, \dots, \varphi(n)$$

$$\text{Therefore: } a^{\varphi(n)} \cdot \prod_{i=1}^{\varphi(n)} r_i = \prod_{i=1}^{\varphi(n)} ar_i = \prod_{i=1}^{\varphi(n)} r_{f(i)} = \prod_{i=1}^{\varphi(n)} r_i \pmod{n}.$$

$$\gcd(r_1, \dots, r_{\varphi(n)}, n) = 1$$

Upon multiplying by a modular inverse of $\prod_{i=1}^{\varphi(n)} r_i$, the congruence

$$a^{\varphi(n)} \equiv \prod_{i=1}^{\varphi(n)} r_i \equiv \prod_{i=1}^{\varphi(n)} r_i \pmod{n}$$

this implies $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Problem 31:

State: Euler's Totient Function:

Euler's Totient Function $\varphi(n)$ counts the positive integers up to a given integer n that are relatively prime to n . It can be defined as the number of integers k in the range $1 \leq k \leq n$ such that $\gcd(n, k) = 1$.

Problem 32:

State Fermat's little theorem: and show how it follows from Euler's totient

Let p be a prime and let a be an integer which is not divisible by p . Then: $a^{p-1} \equiv 1 \pmod{p}$

- p is prime $\rightarrow \varphi(p) = p-1$
- a is not divisible by p $\gcd(a, p) = 1$
- Euler's totient theorem $\rightarrow a^{\varphi(p)} = a^{p-1} \equiv 1 \pmod{p}$

Problem 33:

Describe the Fermat primality test.

Recall, Fermat's little theorem: if p is a prime and $a \in \mathbb{Z}$ with $\gcd(a, p) = 1$ then $a^{p-1} \equiv 1 \pmod{p}$. Hence if $a^{n-1} \not\equiv 1 \pmod{n}$ for $n \geq 1$ and $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$ then n is not prime.

This provides a way of proving that some numbers are composite without constructing a proper divisor.

Problem 34:

Define what is meant by a group.

A group $(G, *)$ consists of a set G and a binary operation (multiplication) $*: G \times G \rightarrow G$, $(g, h) \rightarrow g * h$.

such that the following axioms are satisfied:

- There exists a distinguished identity element $e \in G$ such that for all $g \in G$, we have $e * g = g = g * e$
- For each $g \in G$, there exists a distinguished inverse $\bar{g} \in G$ which satisfies $g * \bar{g} = e = \bar{g} * g$.
- Associativity of $*$: For all $g, h, k \in G$ we have

$$(g * h) * k = g * (h * k).$$

If in addition $g * h = h * g$ for all $g, h \in G$, then the group G is called commutative (abelian).

Problem 35:

Define what it means for a group to be abelian:

If $g * h = h * g$ for all $g, h \in G$ then the group G is abelian.

Problem 36:

Define what it means for a group to be cyclic. Give an example of a cyclic and non-cyclic group.

A group $(G, *)$ is cyclic if there exists an element $g \in G$ with $G = \langle g \rangle$; such an element is called a generator of the group.

Every cyclic group is commutative but not vice versa.

- Cyclic group: $\mathbb{Z}_5^* = \{1, 2, 3, 4\} = \{2^0, 2^1, 2^2, 2^3\} = \langle 2 \rangle_{\mathbb{Z}_5^*}$
- Non cyclic: $\mathbb{Z}_8^* = \{1, 3, 5, 7\}$ has order 4 ($= \varphi(8)$) but $1^2 = 3^2 = 5^2 = 7^2 = 1$ in $\mathbb{Z}_8^*$. Therefore each element has order ≤ 2 and $\mathbb{Z}_8^*$ is not cyclic.

Problem 37:

For which $n \geq 1$ is the group $\mathbb{Z}_n^*$ cyclic:

Theorem: (Gauss): $\mathbb{Z}_n^*$ is cyclic if and only if $n = 1, 2, 4$ or n is of the form $n = p^a$ or $n = 2p^a$ for an odd prime p and integer $a \geq 1$.

Problem 38:

Describe the method of fast exponentiation (modular exponentiation).

Both RSA and the Fermat primality test require raising integers to large powers modulo some number. In practice, we should not require to compute $a^{n-1} \bmod n$ by first determining a^{n-1} . Instead we may use the following algorithm:

- Input: $a \in \mathbb{Z}$, $e, n \geq 1$
- Output: $a^e \bmod n$
- Write $e = e_0 + 2e_1 + \dots + 2^k e_k$, where $e_0, \dots, e_k \in \{0, 1\}$ and $e_k = 1$.
- Use the recursive formula: $a^{2^{j+1}} \bmod n = (a^{2^j} \bmod n)^2 \bmod n$ to compute $a^{2^j} \bmod n$ for $j = 0, \dots, k$
- Determine $a^e \bmod n = \prod_{j: e_j=1} a^{2^j} \bmod n$ again reducing mod n after each multiplication.

Problem 39:

Describe the Diffie-Hellman key exchange protocol:

- Alice and Bob first agree on a finite cyclic group G of order n with

generator g . (Eg. $G = \mathbb{Z}_p^*$, p prime).

- Alice secretly chooses a random integer $a \in \{1, \dots, n-1\}$ she then computes $A := g^a$ in G and sends A to Bob.
- Bob secretly chooses a random integer $b \in \{1, \dots, n-1\}$, computes $B := g^b$ in G and sends B to Alice
- Alice receives B and computes $B^a = g^{ab}$ in G (without knowing b !)
- Bob receives A and computes $A^b = g^{ab}$ in G (without knowing a !)
- Their shared secret key is g^{ab} .

Problem 40:

Alice and Bob use the cyclic group $\mathbb{Z}_{23}^*$ and its generator $g=5$. Alice secretly chooses $a=3$ and Bob secretly chooses $b=4$.

Determine their shared key.

- Alice chooses $a=3$, computes $A = g^a = 5^3 = 125$ and sends it to Bob.
 - Bob chooses $b=4$, computes $B = g^b = 5^4 = 625$ and sends it to Alice
 - Alice receives $B=625$ and computes $B^a = 244140625$
 - Bob receives $A=125$ and computes $A^b = 244140625$
- This is their shared key.

Problem 41:

Describe the key generation step, the encryption step and the decryption step of the ElGamal cryptosystem.

Key Generation:

Alice chooses a cyclic group G of order n and a generator g of G . She chooses a random exponent $a \in \{1, \dots, n-1\}$ and computes

$A := g^a$ in G . Her public key is (G, g, A) and her private key is (G, g, a) .

Encryption:

Bob encrypts a message $m \in G$ as follows. First he chooses a random $b \in \{1, \dots, n-1\}$ and computes $B = g^b$ in G . He then computes $c := A^b * m$. The ciphertext corresponding to the plaintext m is (B, c) .

Decryption:

Alice receives $(B, c) = (g^b, g^{ab} m)$. She computes $B^a = g^{ab}$ and

recovers $m = (B^A)^{-1} * c$

Problem 42:

State Hasse's theorem on the number of points on an elliptic curve over $\mathbb{Z}$.
Hasse's theorem: $|E - p + 1| \leq 2\sqrt{p}$ so the number of points is close to $p + 1$, the number of points on the "projective line" over $\mathbb{Z}_p$.

Problem 43:

Let E be the set of points of an elliptic curve over $\mathbb{C}$ defined by the equation $y^2 = x^3 + ax + b$. Describe the usual operation which turns the set of points of E into an abelian group.

For any elliptic curve $y^2 = x^3 + ax + b$ over a field $\mathbb{F}$ in which $2 \neq 0 \neq 3$, we obtained a commutative group structure on the set E of points of the curve. (The point at infinity is the identity element).